

A Temporal Stress Test of Bankruptcy Prediction Models on U.S. Public Firms

Ayush Bodalia*
Nexus Labs
Palo Alto, USA

Vishruth Burra*
Nexus Labs
Palo Alto, USA
vishruth.burra@gmail.com

Arnav Deokule*
Nexus Labs
Palo Alto, USA
arnavdeokule@gmail.com

Shravan Ginde*
Nexus Labs
Palo Alto, USA
shravanginde1224@gmail.com

Sahasra Namburu*
Nexus Labs
Palo Alto, USA
sahasran0502@gmail.com

Kavindran Rajangam*
Nexus Labs
Palo Alto, USA
kavinrajangam11@gmail.com

Sreehaas Voruganti*
Nexus Labs
Palo Alto, USA
sreehaasvoruganti@gmail.com

Srikanth Samy
University of California, Berkeley
Berkeley, USA
ssamy@berkeley.edu

Ishaan Gupta
Stanford University
Stanford, USA
ishaangupta@stanford.edu

Abstract—Corporate-distress models are typically trained and validated on data pooled across time, yet they are deployed forward into economic conditions that may differ sharply from the training period. We ask whether a bankruptcy classifier trained on calm pre-crisis years still functions during a recession. Using 78,682 company-years of U.S. public firms (1999–2018) with 18 financial variables, we frame imminent-failure prediction—a firm’s final reporting year before delisting—and engineer 15 Altman-style financial ratios. Training on 1999–2006 with a company-grouped split to prevent leakage, a random forest attains a held-out ROC-AUC of 0.755, well ahead of logistic regression (0.646) and a multilayer perceptron (0.625); the distress boundary is strongly nonlinear. Stress-testing the frozen calm-era models on the 2008–2009 recession, the random forest does not degrade but improves to 0.798 AUC, while the linear and neural models slip. The financial-distress signal is thus stable, even sharper, under crisis conditions: the model survives the stress test. We also show that a naive per-year AUC curve suggests a spurious “cliff” at 2007 that is an evaluation artifact of training-set overlap, underscoring the importance of leakage-free temporal validation.

Keywords—bankruptcy prediction, distribution shift, temporal validation, financial ratios, class imbalance

I. Introduction

Models that flag firms at risk of failure are central to credit and investment decisions. Their operational value depends on temporal robustness: a model fit on historical data must keep working as macroeconomic conditions evolve. The 2008 financial crisis is a natural stress test—a sharp regime change between the calm mid-2000s and the recession. We pose a direct question: does a distress model trained on calm years still rank failing firms during the recession, and which model family is most robust?

A. Contributions

- 1) An imminent-failure formulation of the widely used U.S. bankruptcy panel, in which only a firm’s final reporting year counts as positive, paired with company-grouped validation so that no issuer is ever split across a training and an evaluation partition.
- 2) A calm-era baseline across three model families, in which a random forest clearly outperforms linear

and neural alternatives, locating the difficulty of the problem in the nonlinearity of the distress boundary rather than in the feature set.

- 3) A temporal stress test that freezes the calm-era models and applies them to the 2008–2009 recession, finding that the random forest does not merely survive the regime change but improves on it, while the linear and neural models slip.
- 4) A cautionary analysis of a per-year evaluation artifact that can masquerade as a crisis-induced collapse, and which disappears once training-set overlap is removed.

II. Related Work

Distress prediction dates to Beaver’s univariate ratio analysis [1], Altman’s Z-score [2], and Ohlson’s logit model [3], whose ratio families we adopt. Hazard formulations that respect the panel structure of firm data followed [4], [5]. Machine-learning approaches, including random forests [6], are now standard, and tree ensembles remain strong on tabular problems of this shape [7]. The robustness of predictive models under temporal or covariate shift is the subject of the dataset-shift literature [8], and the specific failure mode we document in Section V-C belongs to the literature on leakage in applied machine learning [9], [10]. We combine these threads into an explicit calm-to-recession transfer evaluation on a large panel of U.S. firms.

III. Data

We use a public dataset of 78,682 company-years for U.S. public companies, 1999–2018, each with 18 accounting variables (assets, liabilities, income-statement items, market value) and an alive/failed status. Failure is rare, so plain accuracy is uninformative and we evaluate with ranking and precision-recall metrics (Figure 1). Distressed firms differ systematically from healthy ones—lower profitability and higher leverage—which is the signal our models exploit.

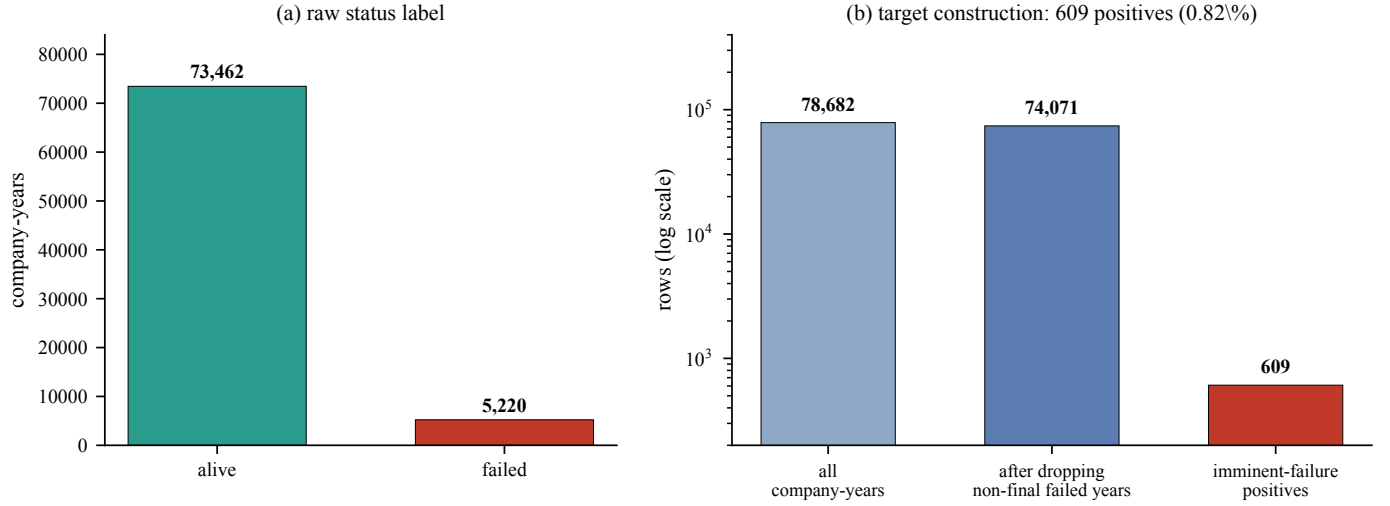


Fig. 1. (a) Raw status labels: 6.6% of company-years belong to firms that eventually failed. (b) The imminent-failure target is far rarer still. Dropping the non-final years of failed firms leaves 74,071 rows, of which only 609 (0.82%) are positive.

A. Target definition

A failed firm appears in the panel for several years, every one of them stamped failed. Labelling all of those years as bankruptcies would be misleading, because a firm five years from delisting usually looks healthy. We therefore label a company-year positive if and only if it is that firm’s final reporting year,

$$y_{it} = \mathbb{1}[\text{status}_i = \text{failed} \wedge t = \max\{t' : (i, t') \in \mathcal{D}\}], \quad (1)$$

and we drop the earlier years of failed firms so that the negative class is clean. This yields 609 positives among 74,071 company-years, a base rate of 0.82%. The label is a proxy: a firm’s last filing is not always the year it legally enters bankruptcy.

B. Features

Raw dollar magnitudes are size-confounded—a \$1M loss is fatal for a small firm and a rounding error for a large one. From the 18 raw variables we therefore build 15 Altman-style ratios [2], [3]: return on assets, EBIT/assets, EBITDA/assets, working-capital/assets, retained-earnings/assets, sales/assets, leverage, long-term-debt/assets, market-value/liabilities, current ratio, gross and net margins, inventory/assets, receivables/assets, and $\log(1 + \text{total assets})$ as a size term. Each is winsorized at the training-set 1st and 99th percentiles, median-imputed, and standardized, with all three statistics estimated on the training rows only:

$$\tilde{x}_j = \frac{\text{clip}(x_j; q_{0.01,j}^{\text{tr}}, q_{0.99,j}^{\text{tr}}) - \mu_j^{\text{tr}}}{\sigma_j^{\text{tr}}}. \quad (2)$$

IV. Methods

A. Models and metrics

Logistic regression predicts $\hat{p}(x) = \sigma(w^\top \tilde{x} + b)$; the random forest [6] averages 400 decision trees with a

minimum of three samples per leaf; the MLP applies two ReLU hidden layers (64 and 32 units) with dropout 0.3 and a sigmoid output, trained with Adam [11] and early stopping. Given the extreme imbalance, all models use class weighting; the MLP minimizes a weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [\beta_1 y_i \log \hat{y}_i + \beta_0 (1 - y_i) \log(1 - \hat{y}_i)],$$

$$\beta_c = \frac{N}{2N_c}. \quad (3)$$

We report ROC-AUC, the probability that a failing firm is ranked above a surviving one [12],

$$\text{AUC} = \Pr(\hat{p}(x^+) > \hat{p}(x^-)), \quad (4)$$

and PR-AUC (average precision), whose no-skill baseline equals the positive rate. With a base rate near 0.6%, PR-AUC is the more demanding summary and the more honest one for a rare-event problem [13], [14].

B. Validation protocol

The calm era (1999–2006) is split by company using a grouped shuffle split, so that a firm never appears in both training and calm-test partitions. This matters because each firm contributes several highly correlated yearly rows; a random row split would let the model recognise a company rather than learn a warning sign. The recession window (2008–2009) is held out entirely for the stress test, and 2007 is excluded as a transition year. Figure 2 summarises the resulting partitions.

We define the calm-to-recession change as

$$\Delta = \text{AUC}_{\text{recession}} - \text{AUC}_{\text{calm}}, \quad (5)$$

where both terms are computed on firms excluded from training. A non-negative Δ means the model survives the regime shift.

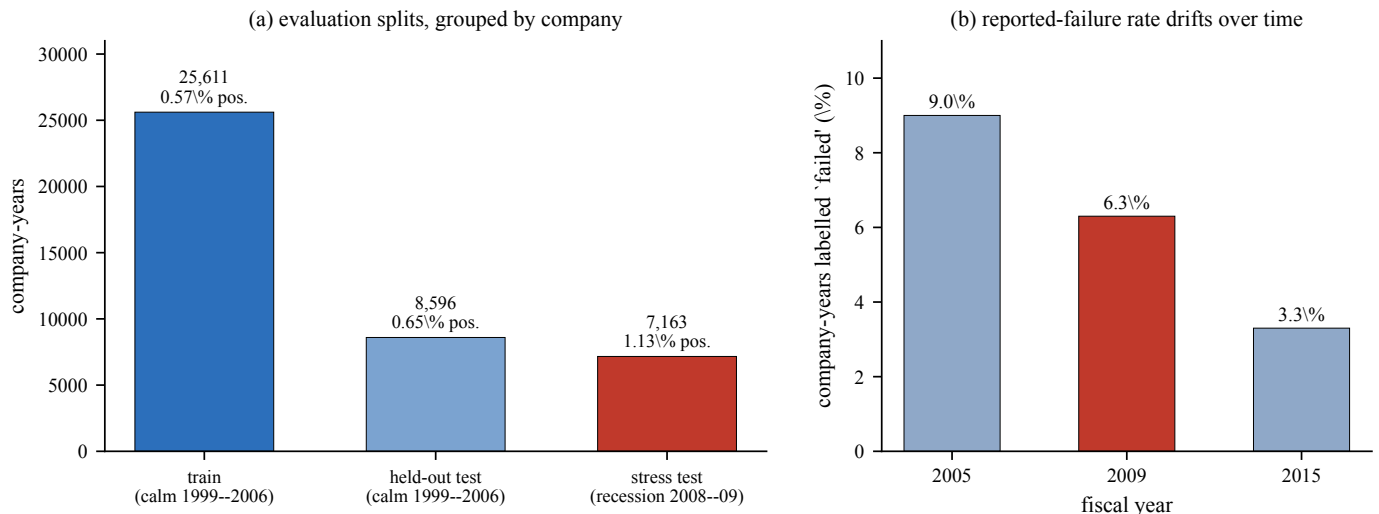


Fig. 2. (a) The three evaluation partitions, with sizes and positive rates. The recession window carries roughly twice the imminent-failure rate of the calm training data (1.13% versus 0.57%). (b) The share of company-years labelled failed drifts substantially over the panel.

V. Experiments and Results

A. Calm-era baseline

Table I reports held-out calm-era performance. The random forest (0.755 AUC) clearly beats logistic regression (0.646) and the MLP (0.625): the mapping from financial ratios to imminent failure is strongly nonlinear and interaction-heavy—low profitability and high leverage and weak liquidity—which the tree ensemble captures and the linear and shallow models do not. PR-AUC is low throughout, an honest consequence of the $\sim 0.6\%$ base rate; even so, the random forest’s 0.020 is roughly $3\times$ the no-skill baseline.

B. The 2008–2009 stress test

Table II and Figure 3 report the frozen calm-era models applied to the recession. The random forest does not degrade; its AUC rises to 0.798 ($\Delta = +0.042$). The linear and neural models slip modestly ($\Delta = -0.031$ and -0.050). Firms about to fail during the crisis are, if anything, easier to identify from their financial ratios than in calm times—distress is a stable, even sharpened, signal. By ranking, the model survives the stress test.

Two points qualify the headline. First, the recession’s higher base rate (1.13% versus 0.57%) raises the attainable precision independently of model quality, so part of the PR-AUC improvement is mechanical rather than evidence of better discrimination; this is why we make ROC-AUC the primary endpoint. Second, the ranking that survives is not a calibrated probability. A threshold tuned on calm-era scores will fire at a different rate when the underlying failure rate roughly doubles, so operating points must be re-calibrated even though the ordering is preserved [15].

C. A per-year artifact that looks like a crisis break

It is natural to diagnose temporal robustness by scoring the calm-trained model on every fiscal year and plotting

the resulting AUC curve. Doing so produces an apparent “cliff” after 2006 that invites the reading that the crisis broke the model. It is largely an evaluation artifact.

The mechanism is straightforward. Pre-2007 years still contain many of the exact firms the model trained on, and because a firm’s yearly rows are highly correlated, scoring those years measures partial memorisation rather than prediction. Later years contain genuinely unseen firms. The apparent drop is therefore not the crisis arriving; it is the evaluation finally becoming honest. This is a textbook instance of the leakage patterns catalogued in [9], [10]: the contaminated estimate is not merely optimistic, it induces a qualitatively wrong conclusion about when performance changed.

The leakage-free comparison shows no break at all. Figure 4 places the two valid estimates side by side: held-out calm firms at 0.755 and held-out recession firms at 0.798. The correct reading of the same underlying model is that performance rose across the regime change.

The practical rule follows directly: in panel data, temporal validation must exclude a firm’s entire history from the test period, not merely the test year.

VI. Discussion

The headline is reassuring and slightly counterintuitive: a distress model built before the crisis survives the crisis, and the most flexible model is also the most robust. Because financial ratios encode the mechanics of insolvency—eroding profitability, mounting leverage, shrinking liquidity—the calm-era decision surface transfers to the recession, where these signatures are sharper.

That the random forest is simultaneously the most accurate and the most robust deserves comment, because flexibility and robustness are often assumed to trade off. Here the distress boundary is genuinely nonlinear, so the linear model is misspecified in both regimes and has no

TABLE I
Held-out calm-era (1999–2006) performance, company-grouped split. The PR-AUC no-skill baseline is the positive rate, 0.0065.

Model	ROC-AUC	PR-AUC
Logistic Regression	0.646	0.010
Random Forest	0.755	0.020
MLP	0.625	0.012

TABLE II
Calm-era versus 2008–2009 recession, both on firms excluded from training. Δ is defined in Eq. (5).

Model	Calm AUC	Recession AUC	Δ
Logistic Regression	0.646	0.615	−0.031
Random Forest	0.755	0.798	+0.042
MLP	0.625	0.575	−0.050

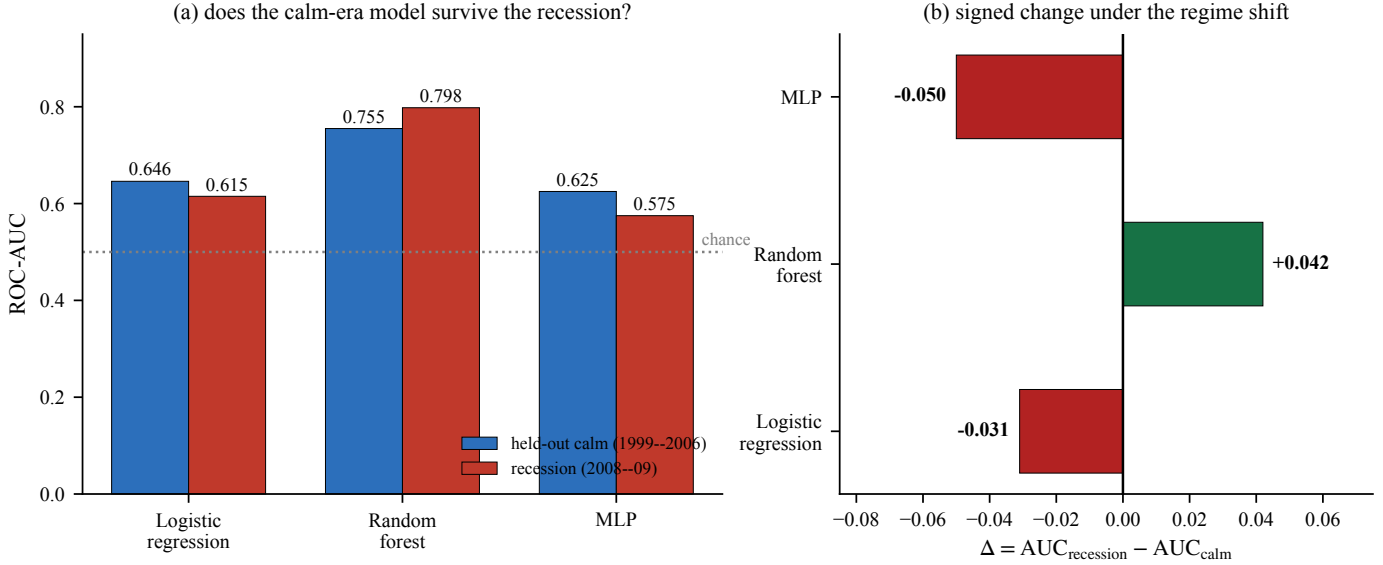


Fig. 3. (a) Calm versus recession ROC-AUC per model. Taller red than blue means the model holds up. (b) The signed change of Eq. (5): only the random forest is positive, and it is the only model that was strong to begin with.

robustness advantage to offer; its smaller absolute loss under shift comes with a much lower absolute score. The MLP, fitting ~ 150 positives, is data-starved rather than over-flexible. The practical caveats concern calibration and measurement, not ranking.

VII. Limitations

Positives are rare, so PR-AUC is low and the recession estimates rest on a few dozen failures; they should be read as directional. The imminent-failure label—the final reporting year—is a proxy for the legal bankruptcy date. Firms recur across the calm and recession windows, so the transfer is across time rather than to entirely new firms; this is realistic for re-scoring a portfolio but is a caveat for generalization to unseen issuers. We report a single seed and a single design, with no confidence intervals, so differences smaller than roughly 0.02 AUC should not be interpreted. Finally, survivorship and reporting biases affect which firms appear in the panel at all.

VIII. Model Card

Table III states the intended and prohibited uses of the calm-era random forest in the format of [16]. Every entry restates a decision already made in the sections above; it is collected here because the distinction between a ranking that survives a regime change and an operating point that does not is exactly the distinction a deployment reader needs.

IX. Conclusion

On a large panel of U.S. public firms, a random forest trained on calm pre-crisis years predicts imminent bankruptcy at 0.755 ROC-AUC and survives the 2008–2009 recession, improving to 0.798, while simpler models slip. Financial-distress signals are temporally robust, and a per-year evaluation artifact can falsely suggest a crisis-induced collapse. We recommend leakage-free temporal validation and threshold re-calibration for deployment across economic regimes.

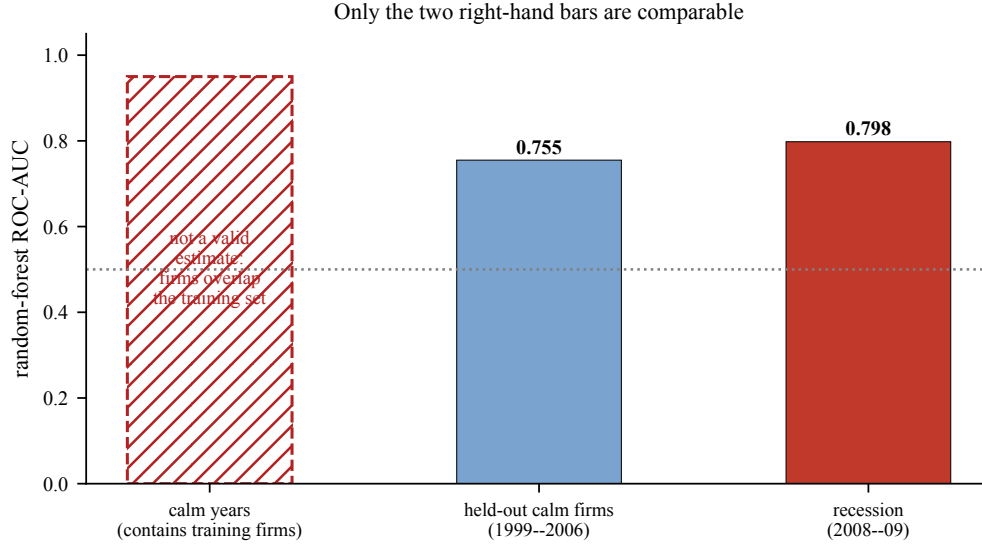


Fig. 4. Only estimates computed on firms excluded from training are comparable. A per-year curve scored on calm years is contaminated by training-firm overlap (left, hatched) and cannot be placed on the same axis as the two valid estimates.

TABLE III
Model card [16] for the calm-era random forest.

Item	Statement
Intended use	Research-grade ranking of U.S. public company-years by imminent-failure risk.
Training population	Calm-era company-years, 1999–2006, split by company.
Evaluation population	Held-out calm-era firms and the 2008–2009 recession window; 2007 is excluded as a transition year.
Positive class	A firm’s final reporting year before delisting, Eq. (1); base rate 0.82%.
Features	The 15 Altman-style ratios of Equation (2), winsorized, median-imputed and standardized on training rows only.
Primary metric	ROC-AUC. PR-AUC is reported alongside and is the more demanding summary at this base rate.
Decision threshold	None. The model is evaluated purely by ranking; no operating point is fitted or recommended.
Known weak point	Scores are not calibrated probabilities. The recession base rate roughly doubles, so a threshold set on calm-era scores fires at the wrong rate and must be re-calibrated.
Prohibited use	Credit decisions about individual firms, regulatory capital calculations, or any use that reads the score as a probability of bankruptcy.

Reproducibility

The analysis notebook fixes all random seeds, defines the target and the company-grouped split explicitly, and estimates every preprocessing statistic inside the training partition. The source panel is licensed and is not redistributed here; the figure script accompanying this manuscript regenerates all figures and tables from the recorded results, and recomputes the per-year analysis when the raw CSV is supplied.

References

- [1] W. H. Beaver, “Financial ratios as predictors of failure,” *Journal of Accounting Research*, vol. 4, pp. 71–111, 1966.
- [2] E. I. Altman, “Financial ratios, discriminant analysis and the prediction of corporate bankruptcy,” *The Journal of Finance*, vol. 23, no. 4, pp. 589–609, 1968.
- [3] J. A. Ohlson, “Financial ratios and the probabilistic prediction of bankruptcy,” *Journal of Accounting Research*, vol. 18, no. 1, pp. 109–131, 1980.
- [4] T. Shumway, “Forecasting bankruptcy more accurately: A simple hazard model,” *The Journal of Business*, vol. 74, no. 1, pp. 101–124, 2001.
- [5] J. Y. Campbell, J. Hilscher, and J. Szilagyi, “In search of distress risk,” *The Journal of Finance*, vol. 63, no. 6, pp. 2899–2939, 2008.
- [6] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [7] L. Grinsztajn, E. Oyallon, and G. Varoquaux, “Why do tree-based models still outperform deep learning on typical tabular data?” *Advances in Neural Information Processing Systems*, vol. 35, pp. 507–520, 2022.
- [8] J. Quiñonero-Candela, M. Sugiyama, A. Schwaighofer, and N. D. Lawrence, *Dataset Shift in Machine Learning*. Cambridge, MA: MIT Press, 2009.
- [9] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, “Leakage in data mining: Formulation, detection, and avoidance,” *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 1–21, 2012.
- [10] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, p. 100804, 2023.
- [11] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Rep-*

resentations (ICLR), 2015.

- [12] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.
- [13] J. Davis and M. Goadrich, "The relationship between precision–recall and ROC curves," in *Proceedings of the 23rd International Conference on Machine Learning (ICML)*, 2006, pp. 233–240.
- [14] T. Saito and M. Rehmsmeier, "The precision–recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLoS ONE*, vol. 10, no. 3, p. e0118432, 2015.
- [15] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, and E. W. Steyerberg, "Calibration: The Achilles heel of predictive analytics," *BMC Medicine*, vol. 17, no. 1, p. 230, 2019.
- [16] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, "Model cards for model reporting," *Proceedings of the Conference on Fairness, Accountability, and Transparency*, pp. 220–229, 2019.